Mystery unraveled: Citrus Greening Causal Organism, *Candidatus Liberibacter asiaticus* Detected in Bangladeshi Sweet Orange

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Background

The gram negative fastidious unculturable bacteria *Candidatus Liberibacter asiaticus* (Las) is the causal agent of citrus greening (huanglongbing; HLB). It is a destructive pathogen and widespread in Asia as well as in the Western hemisphere (Bové, 2006). It is spread by grafting and sap-sucking insect vector Asian citrus psyllid (*Diaphorina citri* Kuwayama) (Lin, 1963; Martinez & Wallace, 1967). Detection of Las is not easy because of its low concentration and irregular distribution in its citrus hosts (McClean, 1970). This disease is also difficult to be diagnosed by conventional procedures, whereas polymerase chain reaction (PCR) is the most convenient method for detection (Jagoueix *et al*., 1996).

Fig. 1. Symptom of Citrus greening showing yellowing of leaves

Fig. 2. Asian citrus psyllid was found in citrus orchard

Fig. 3. Leaf symptoms of (a) citrus greening showing yellowing and mottling of leaf, (b) healthy leaf

Fig. 4. Map of sampling sites in Bangladesh for *Ca. L. asiaticus* detection in citrus

Objectives

To date, there is no report available about diagnosis or detection of citrus greening causal organism by molecular technique in Bangladesh. Therefore, the study was undertaken to detect and identify the citrus greening causal organism, *Candidatus Liberibacter asiaticus* by PCR and sequencing.

Materials and Methods

- In July 2016, suspected leaves with characteristic irregular blotchy chlorosis symptom were collected from major citrus growing areas of Bangladesh.
- The Las genomic DNA was extracted from midribs and petioles of symptomatic leaf using Wizard® Genomic DNA Purification Kit (Promega, Madison, USA) following manufacturer's instructions.
- The samples were then examined for the detection of *Candidatus Liberibacter asiaticus* by DNA-based polymerase chain reaction (PCR) using the 'Las' specific primers "A2" and "J5" to amplify a partial sequence of β-operon (*rpl*KAJL-*rpo*BC operon) of ribosomal protein genes (Hocquellet *et al*. 1999).
- PCR products were sequenced directly in both orientations from primers A2/J5 according to the standard protocols for the ABI PRISM 3500 (version 3.7).

Results

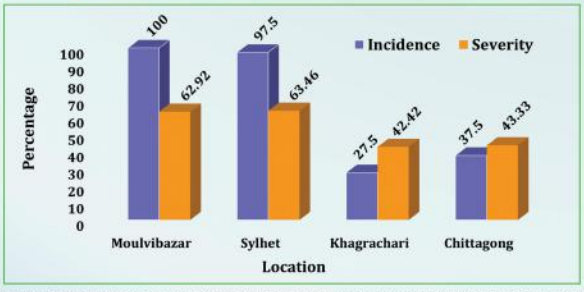

Fig. 5. Incidence and severity of citrus greening in different geographic location of Bangladesh

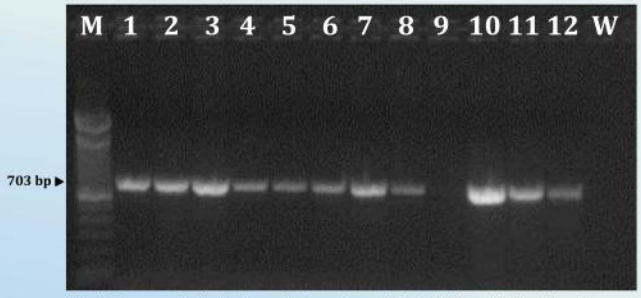

Fig. 6. Agarose gel electrophoresis of symptomatic leaf DNA amplified with primer pair A2/J5, specific for *Candidatus Liberibacter spp*. Lane M: 100 bp DNA ladder; lanes 1 to 3, DNA from "*Ca. Liberibacter asiaticus*" infected leaves from Khagrachari, lanes 4 to 6 DNA from "*Ca. Liberibacter asiaticus*" infected leaves from Moulvibazar, lanes 7 to 9 DNA from "*Ca. Liberibacter asiaticus*" infected leaves from Sylhet, lanes 10 to 12 DNA from "*Ca. Liberibacter asiaticus*" infected leaves from Hathazari and lane W: Water.

- A band of about 703 bp specific to *Ca. L. asiaticus* strains was obtained after PCR amplification.
- Partial nucleotide sequences of ribosomal protein genes of *rpl*KAJL-*rpo*BC operon showed 99% sequence identity with *Ca. L. asiaticus* strains found in India, China, Indonesia, and USA (GenBank accession # JQ964017, DQ157278, LC090233, and KR919751, respectively).
- In case of amino acid level it showed 97-98% protein sequence identity with different isolates deposited in GenBank (Accession # ACY26056, ABI35994, etc.).

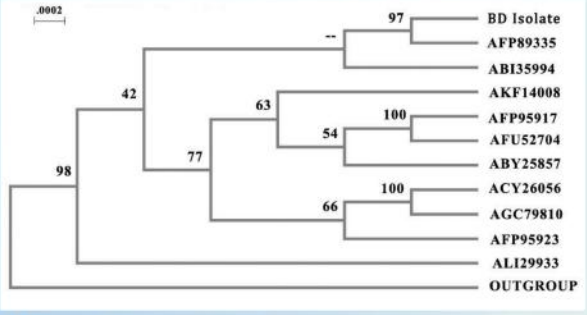

Fig. 7. Phylogenetic trees based translated amino acid sequences of ribosomal protein genes of β-operon showing the positions of Bangladeshi isolate and representatives of *Candidatus liberibacter* spp. and strains available in the Genbank.

- Phylogenetic tree showed close relationship with Indian and Thai isolates.

Summary and Conclusion

- The origin of *Ca. Liberibacter asiaticus* in Bangladesh is uncertain.
- We believe that it probably was introduced by infected material from South Asian region.
- Results from PCR amplification and aligning ribosomal gene sequences gave strong evidence that the *Ca. L. asiaticus* is present in Bangladesh.
- It revealed the mystery of citrus declining in Bangladesh which was associated with bacterial pathogen and not due to micronutrient deficiencies.
- This is the first molecular detection of citrus greening causal *Candidatus Liberibacter asiaticus* in the country.

References

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Acknowledgments

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